Milestone 1: PanePlan - Glass Measure

- ## 1. Problem and User

  - **User:** Field installers (handling aluminum and shutters) and technical managers who prepare quotes/invoices in the office.
  - **Problem:** Measurements recorded on paper at the site often get lost, arrive late at the office, or are misread, causing delays in the quotation process.
  - **Evidence/Interview:** In a team coordination meeting, it was identified that transferring paper measurements to digital formats causes significant time loss and miscommunication between the field and the office.

2. Why Important

Existing standard calculators do not save data or offer historical tracking. Comprehensive ERP systems available on the market are too complex for field workers and require a constant internet connection. There is a need for a practical, mobile solution tailored for the field that can work offline, save measurements to local storage, and automatically sync to the cloud when an internet connection is established.

3. Features (mapped to course topics)

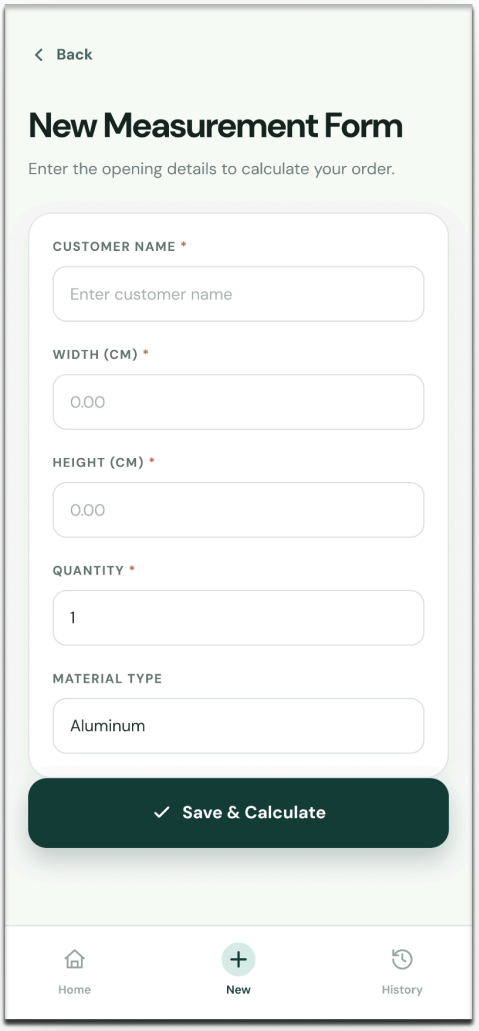
- **New Measurement Form:** Allows field staff to enter dimensions and save them locally without internet (Local DB).
- **Recent Projects List:** Displays saved jobs and allows navigation to detailed view pages (Navigation).
- **Cost Calculation Module:** Separates the user interface from the pricing logic (MVVM).
- **Cloud Synchronization:** Transfers field data to the cloud for office access once an internet connection is available (Firebase).
- **User Feedback:** Provides confirmation alerts upon saving and ensures UI fluidity (Animation).

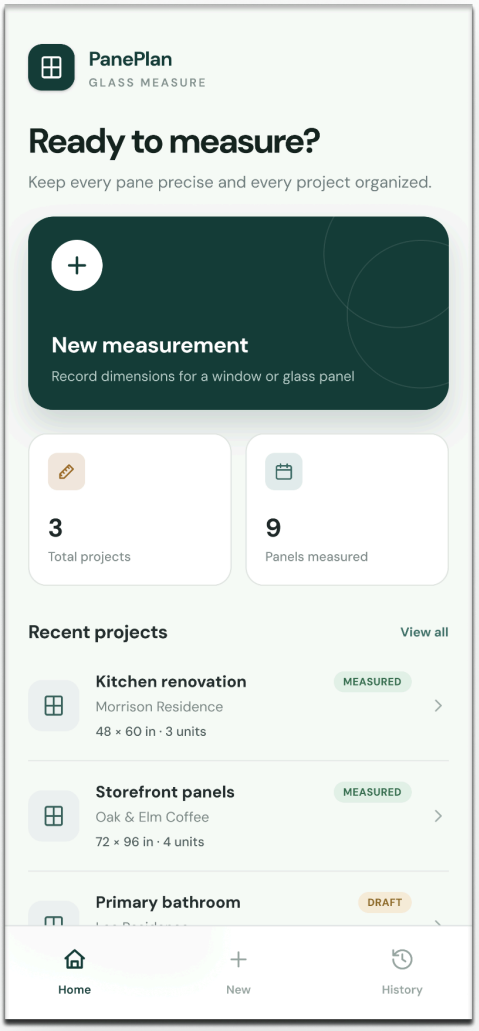
- ## 4. Architecture

  - **Screens Map:** Home (Ready to measure?) → New Measurement Form → Recent Projects → Order Details.
  - **Data Model:** Project(id, customer_name, width, height, material_type, total_price, date).
  - **State Management:** Riverpod (or Provider).

- ## 5. Plan and Risks
    - **Weekly Tasks:** Distribute UI Development, Local Database, Firebase Integration, and Testing processes among team members.
    - **Risks:** Potential data conflicts during offline synchronization and possible delays in the Flutter/Dart learning curve.

6. Mockups


Back
New Measurement Form
Enter the opening details to calculate your order.
CUSTOMER NAME *
Enter customer name
WIDTH (CM) *
0.00
HEIGHT (CM) *
0.00
QUANTITY *
1
MATERIAL TYPE
Aluminum
Save & Calculate
Home
New
History


PanePlan
GLASS MEASURE
Ready to measure?
Keep every pane precise and every project organized.
New measurement
Record dimensions for a window or glass panel
3
Total projects
9
Panels measured
Recent projects
View all
Kitchen renovation
MEASURED
Morrison Residence
48 × 60 in · 3 units
Storefront panels
MEASURED
Oak & Elm Coffee
72 × 96 in · 4 units
Primary bathroom
DRAFT
Home
New
History